TOPIC 03

Graphical Representation of Regression

The regression line and its parts, a first worked example with SAT scores and college GPA, and how to use the equation for prediction

WHAT YOU WILL LEARN

- Write the simple linear regression equation and name b_0, b_1, $\hat{y}$ and the residual
- Draw and read a regression line on a scatter plot
- Interpret the slope and the intercept in words
- Decide which variable is x and which is y
- Predict y for a given x, and know when not to
- See how rounding the coefficients changes predictions

NOTE **Where this connects**

Topic 02 contrasted correlation (a single number) with regression (a line with a direction). Here we draw that line and read it. The same SAT and GPA data also show why correlation alone is not enough: we first confirm that the two variables are correlated (and test it, as in the p-value topic), and only then fit the line.

1. The linear regression model

$$\hat{y}_i = b_0 + b_1 x_i$$

Symbol	Name	Meaning
x_i	Observed value of x (independent variable)	The predictor, e.g. SAT score
y_i	Observed value of y (dependent variable)	What actually happened, e.g. GPA
$\hat{y}_i$	Predicted (fitted) value	The y the line gives for x_i
b_0	**Intercept**	Where the line crosses the y-axis (the value of $\hat{y}$ when x = 0)
b_1	**Slope**	Change in $\hat{y}$ for a **one-unit increase in x**
$e_i = y_i - \hat{y}_i$	**Residual** (estimator of the error)	The gap between the observed point and the line

Reading a line from two points. For y = 2 + 3x, the points (0, 2), (1, 5) and (2, 8) lie on the line. The intercept $b_0 = 2$ is the height where x = 0. The slope $b_1 = 3$ is rise over run: going from x = 0 to x = 1 (run = 1) the line rises from 2 to 5 (rise = 3). Every extra unit of x adds 3 to $\hat{y}$. A fitted line is read exactly the same way, with b_0 and b_1 chosen by least squares.

The residual e_i is the **observable** gap for each student; the unknown true error ε in the population model is what e_i estimates.

The **regression line** is the best-fitting line through the data points: the grey dots in the lecture slide are the observed values, and the orange line is $\hat{y}$. The next topic shows exactly what 'best' means (least squares).

> **KEY IDEA** **Residual: vertical, not perpendicular**
> The lecture text describes dropping a 'perpendicular' from the point to the line. In the slides the dashed segment is **vertical**: the residual is measured parallel to the y-axis, $y - \hat{y}$. That vertical distance is what regression minimises.

In the population the true line is $y = \beta_0 + \beta_1 x + \varepsilon$, where ε is the error. The sample line with b_0 and b_1 is our **estimate** of it, in the same way $\bar{x}$ estimates μ.

2. Worked example: SAT score and college GPA

The data are 84 students. **SAT** is the total of the critical reading, mathematics and writing scores. **GPA** is the grade point average at graduation from college. The question: can SAT scores predict college GPA?

1. **Look first.** The scatter plot shows the points drifting upward: a higher SAT tends to go with a higher GPA.
2. **Measure.** Sample correlation $r =$ **0.6372**, a moderately strong positive relationship.
3. **Test it** (this is hypothesis testing, Topics 10 and 13). H_0: $\rho = 0$. The test statistic is $t = r\sqrt{(n-2)}/\sqrt{(1-r^2)} =$ **7.49** with 82 degrees of freedom, giving p = 7.2e-11. We reject H_0: the two variables really are correlated.
4. **Fit the line** (details in Topic 04).

Note what correlation does **not** give us: no information about under- or over-achievers, and no ability to predict a GPA for a particular student. For that we need the regression line.

3. Which variable is x?

The data could be plotted either way, giving two regression lines (Topic 02). We must choose one. The decision comes from logic, not from the numbers.

Sequence of events	Variable
1. A student takes the SAT	SAT score: **x** (independent)
2. The score is used to apply to college	—
3. Over the next four years the student earns grades	College GPA: **y** (dependent)

SAT comes first and GPA later, so SAT can influence GPA but not the other way round. The reversed regression (SAT on GPA) is **meaningless as a causal story**, even though it is mathematically valid. Colleges also use the SAT as a first part of admission, which gives a plausible **rationale**: it is considered an estimate of intellectual capacity and capability.

> **COMMON MISTAKE** **Think before you regress**
> Finding data that looks 'regressable' is not a licence to run a regression. First ask whether a sensible cause-and-effect direction exists (Topic 01), and only then fit.

4. The fitted line

The least-squares fit to these 84 students gives

$$\widehat{GPA} = 0.2750 + 0.001656 \times SAT$$

- **Slope b_1 = 0.001656**: each extra SAT point goes with about 0.0017 more GPA points. A 100-point higher SAT goes with about **0.17** more GPA points.
- **Intercept b_0 = 0.2750**: the predicted GPA for SAT = 0. No student scored near 0 (SAT ranges from 1634 to 2050), so the intercept is just the number that positions the line, not something to interpret literally.

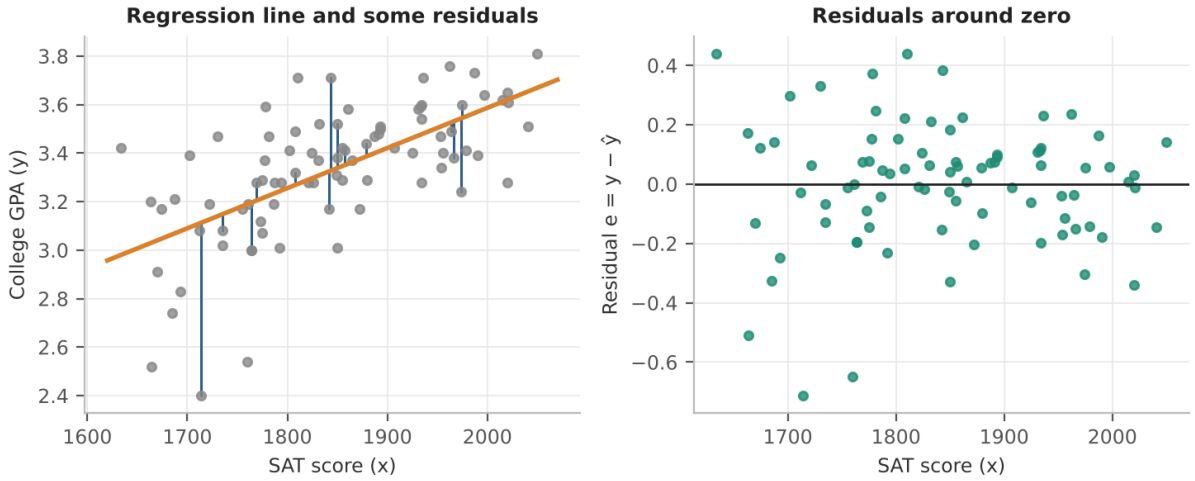


Figure 1. Left: the fitted line with vertical residual segments for every sixth student. Right: the residuals scatter around zero (their sum is −7.9e-14, i.e. zero up to rounding).

5. Using the equation to predict

Insert an SAT score into the equation. The lecture does this for 1,700 and 2,000 using its equation $\hat{y} = 0.48 + 0.0015 \times SAT$.

SAT	Lecture slide: 0.48 + 0.0015x	Lecture workbook, more decimals: 0.4803 + 0.0015492x	Fit to the 84 students
1700	3.03	3.11	3.09
2000	3.48	3.58	3.59

COMMON MISTAKE **Round coefficients with care**

At SAT = 1,700 the slide's rounded slope (0.0015) predicts 3.03, but the unrounded slope from the same regression table gives 3.11. Rounding the slope by only 0.0000492 moves the prediction by 0.0000492 × 1,700 = 0.08 GPA points. Keep several significant figures in b_1 until the final step.

The lecture's reading: a student admitted with an SAT of 1,700 is predicted to graduate with a GPA of about 3.0 to 3.1; one admitted with 2,000, about 3.5 to 3.6. Our own fit to the data gives 3.09 and 3.59.

Prediction is not certainty, and extrapolation is risky

- The line gives an **expected** GPA. Real students scatter above and below it by about one **standard error of the regression** (0.21 GPA points here, Topic 05).
- Predict only within the **range of the data** (SAT about 1,630 to 2,050). An SAT of 2,400 or 800 lies outside what the data support, so the straight-line assumption may fail there.

6. In Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

data = np.loadtxt("satgpa.csv", delimiter=",", skiprows=1)
sat, gpa = data[:, 0], data[:, 1]

fit = stats.linregress(sat, gpa)
print(fit.intercept, fit.slope, fit.rvalue, fit.pvalue)

pred = fit.intercept + fit.slope * np.array([1700, 2000])
print(pred.round(2))                          # [3.09 3.59]

plt.scatter(sat, gpa)
plt.plot(sat, fit.intercept + fit.slope * sat)
plt.xlabel("SAT"); plt.ylabel("GPA"); plt.show()
```

SUMMARY

- Regression line: $\hat{y} = b_0 + b_1 x$, the best-fitting line through the points.
- b_1 is the change in $\hat{y}$ per one unit of x; b_0 is the value at $x = 0$ (often not meaningful).
- Residual $e = y - \hat{y}$ is the **vertical** gap between a point and the line.
- Choose x and y by logic and timing: SAT first, so SAT is x and GPA is y.
- Test that the correlation is real before fitting, then predict; stay inside the data range and keep enough decimals.

Practice questions

Try each question before reading the answer underneath it.

Q1. In $\hat{y} = 2 + 3x$, what does 3 mean?

Answer. Each one-unit increase in x goes with a 3-unit increase in the predicted y.

Q2. Using GPA = 0.2750 + 0.001656 × SAT, predict the GPA for SAT = 1,900.

Answer. 0.2750 + 0.001656 × 1,900 = 3.42.

Q3. A student with SAT 1,900 has an actual GPA of 3.60. Find the residual.

Answer. $\hat{y} = 3.42$, so $e = 3.60 - 3.42 = 0.18$ (the student did better than predicted).

Q4. Why is regressing SAT on GPA a poor causal model?

Answer. GPA is earned years after the SAT is taken, so GPA cannot cause the earlier SAT score.

Q5. Why should you not use this line to predict GPA for an SAT of 900?

Answer. 900 is far outside the observed SAT range (about 1,630–2,050); the straight-line relationship is not supported there.